ICU Mortality Prediction Using Machine Learning: A Practical Framework for Risk Stratification

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***Abstract*—Mortality prediction in intensive care units (ICUs) remains an important clinical challenge because critically ill patients may deteriorate rapidly and require timely intervention. Although conventional severity scoring systems such as APACHE and SOFA are widely used, they may not fully capture complex relationships among patient characteristics, treatment requirements, and clinical instability. This study developed a data science framework for ICU mortality prediction using a public structured dataset containing 15,000 patient records. The workflow included data preprocessing, exploratory analysis, feature refinement, class balancing, and comparative evaluation of multiple machine learning models. Seven clinically relevant predictors were retained for final modeling: ventilation requirement, vasopressor use, sepsis status, APACHE score, SOFA score, comorbidity score, and age. Among the evaluated models, a tuned XGBoost classifier with a decision threshold of 0.35 achieved the most suitable performance for the study objective, obtaining a recall of 0.885, an F1-score of 0.678, and a ROC-AUC of 0.641. Feature importance analysis identified vasopressor use, ventilation requirement, and sepsis as the strongest mortality indicators. These findings suggest that recall-focused machine learning models can provide practical and interpretable support for ICU risk screening and patient prioritization.**

***Index Terms*—ICU mortality prediction, machine learning, XGBoost, clinical data, healthcare analytics, risk stratification**

I. Introduction

Intensive care units (ICUs) are responsible for managing patients with severe and life-threatening conditions who often require continuous monitoring, rapid intervention, and complex treatment decisions. In this environment, patient status can worsen within a short period of time, making early recognition of high-risk cases especially important. Reliable mortality assessment may help clinicians prioritize monitoring intensity, initiate timely interventions, and use limited ICU resources more effectively [12].

To support this need, hospitals have long relied on clinical severity scoring systems such as the Acute Physiology and Chronic Health Evaluation (APACHE) and the Sequential Organ Failure Assessment (SOFA). These tools remain valuable because they provide standardized measures of illness severity and organ dysfunction. However, they are based on predefined scoring rules and may not fully reflect the complex interactions among demographics, comorbidities, treatment requirements, and changing physiological conditions [12].

Recent advances in data science and machine learning have created new opportunities for outcome prediction in critical care. Rather than depending only on fixed formulas, machine learning models can learn patterns directly from historical patient data and evaluate multiple predictors simultaneously. Esteva et al. [6] noted that machine learning methods are becoming increasingly important in clinical outcome prediction because they can identify relationships that are difficult to model using conventional statistical approaches. This is particularly relevant in ICU settings, where outcomes may be influenced by age, chronic disease burden, infection status, hemodynamic instability, and the need for advanced support such as mechanical ventilation or vasopressor therapy.

Recent ICU studies have reported encouraging findings, although their settings and objectives differ. Kim et al. [4] applied machine learning models to predict both ICU mortality and length of stay, while Li et al. [7] used the MIMIC-IV database to estimate in-hospital mortality among ICU patients following cardiac arrest. Zhang et al. [3] further showed that combining dynamic SOFA score changes with age improved 28-day mortality prediction. Together, these studies suggest that data-driven models can complement traditional scoring systems and strengthen clinical risk estimation.

At the same time, strong predictive performance alone does not solve every practical challenge. ICU datasets often contain class imbalance, where survivor cases disproportionately outnumber mortality cases, which can bias models toward the majority class. In addition, highly dimensional datasets may include redundant predictors that add complexity without improving usefulness. Cabitza et al. [10] emphasized that implementation barriers in critical care include data quality concerns, limited generalizability, and the need for inter-

pretable systems that clinicians can trust.

Motivated by these challenges, this paper develops a data science framework for ICU mortality prediction using a structured public dataset containing 15,000 patient records. The workflow includes data preprocessing, exploratory analysis, feature refinement, class balancing, comparative model evaluation, and decision-threshold tuning. Rather than emphasizing accuracy alone, greater importance is given to clinically useful detection of high-risk patients through recall-focused modeling. The objective is to demonstrate how interpretable machine learning methods can support ICU risk screening and future decision-support applications [9], [12].

II. Project Focus and Objectives

This study focuses on developing and evaluating a practical data science framework for predicting mortality risk among ICU patients using structured clinical data. Rather than relying only on traditional severity scores or a single modeling technique, the proposed approach combines preprocessing, feature refinement, class balancing, model comparison, and threshold tuning within one predictive workflow. In addition to predictive performance, emphasis is placed on interpretability and clinically actionable outcomes.

A further objective is to identify a smaller set of relevant predictors from the original dataset. ICU data may contain variables that are weakly associated with outcomes or redundant with other measurements. Reducing unnecessary features can simplify model development, improve transparency, and support clearer interpretation of results. Class imbalance was also treated as an important challenge, since survival cases significantly outnumber mortality cases in many healthcare datasets. Without correction, models may become biased toward the majority class and fail to detect enough high-risk patients [10].

The main objectives of this study are summarized as follows:

1) To analyze a public ICU dataset and identify important clinical patterns associated with mortality risk.
2) To refine the available predictors into a smaller and more informative feature set using statistical and practical considerations.
3) To train and compare multiple machine learning models for ICU mortality prediction.
4) To improve mortality detection through threshold tuning with greater emphasis on recall-sensitive classification.
5) To interpret model outputs using feature importance analysis and risk stratification results.
6) To demonstrate how data science methods can support future ICU decision-support systems [9], [12].

III. Literature Review

A. ICU Mortality Prediction

Predicting mortality in intensive care units is one of the most widely studied applications of machine learning in critical care. ICU environments generate large volumes of structured data, including demographics, severity scores, physiological indicators, laboratory values, and treatment variables. This has encouraged the development of predictive models that can estimate mortality risk more flexibly than fixed rule-based scoring systems alone.

Recent studies suggest that machine learning models can support ICU mortality prediction across different clinical settings. Kim et al. [4] showed that structured hospital data can be used not only for mortality prediction but also for operational outcomes such as length of stay, highlighting the broader value of predictive analytics in intensive care. Li et al. [7] demonstrated similar potential in a specialized subgroup of ICU patients admitted after cardiac arrest, indicating that machine learning approaches may remain useful even when mortality drivers differ by diagnosis.

Another important direction in the literature is the use of dynamic rather than static predictors. Zhang et al. [3] reported improved 28-day mortality prediction when temporal changes in SOFA score were combined with age, suggesting that patient risk should be viewed as evolving throughout hospitalization rather than fixed at admission. During the COVID-19 period, Fadlallah et al. [2] further showed that ensemble approaches such as SuperLearner could achieve strong predictive performance under highly complex ICU conditions.

Despite these advances, several practical limitations remain. Many published models rely on large feature sets, disease-specific cohorts, or performance metrics alone without sufficient attention to interpretability and deployment practicality. Results are also difficult to compare directly because datasets, mortality definitions, and evaluation strategies exhibit considerable heterogeneity across studies.

These findings indicate that ICU mortality prediction remains an active research area where practical, transparent, and reproducible modeling approaches are still needed. The present study contributes to this area through feature refinement, class balancing, threshold tuning, and interpretable analysis using publicly available structured ICU data.

B. Early Warning Systems

While mortality prediction focuses on estimating overall risk outcomes, early warning systems address a closely related objective: identifying clinical deterioration early enough for timely intervention. In ICU and hospital environments, delays in recognizing worsening patient status may reduce treatment opportunities and increase the risk of severe complications. For this reason, early warning models have become an important branch of predictive analytics in healthcare, particularly when built from routinely collected clinical data [12].

A key question in this area is whether predictive systems can remain reliable across different institutions rather than only the hospital in which they were developed. Machado et al. [1] designed a multicenter machine learning early warning system using data from Brazilian hospitals and demonstrated successful external validation across multiple sites. This is important because many healthcare models perform well locally

but lose accuracy when applied to different patient populations, workflows, or documentation practices.

Early warning research is also strongly connected to sepsis detection, since sepsis remains one of the leading causes of rapid deterioration and mortality in critical care. Shashikumar et al. [5], in a systematic review, found growing evidence that machine learning models can identify sepsis risk earlier than conventional recognition methods when electronic health record data are used effectively. Liu et al. [8] similarly applied machine learning techniques to predict mortality among ICU patients with sepsis, reinforcing the value of earlier risk stratification in severe infectious conditions.

Beyond specific diagnoses, early warning systems reflect a broader shift from reactive care toward proactive risk management. Instead of waiting for obvious clinical decline, hospitals can use predictive tools to prioritize monitoring, direct staff attention, and trigger escalation pathways earlier. This is especially valuable in ICUs, where delayed response may carry serious consequences and resources often need to be allocated quickly [6].

However, practical deployment remains challenging. Many early warning systems face issues related to false alarms, alert fatigue, inconsistent data quality, and clinician trust. Cabitza et al. [10] noted that even technically strong models may struggle in practice when outputs are difficult to interpret or poorly integrated into routine workflows. Strong predictive performance alone does not guarantee clinical usefulness.

These findings show that effective early warning systems must balance predictive accuracy with usability in real clinical settings. False alarms, poor integration, and limited interpretability can reduce practical value even when technical performance is strong.

C. High-Risk ICU Conditions

Certain ICU conditions are consistently associated with higher mortality risk because they reflect severe physiological instability, organ dysfunction, or the need for aggressive life-support measures. For this reason, many critical care prediction studies place strong emphasis on variables linked to shock, respiratory failure, infection, and overall disease burden rather than relying on demographic factors alone.

One of the clearest indicators of severe illness is the requirement for mechanical ventilation. Patients requiring ventilatory support are often experiencing respiratory compromise, impaired gas exchange, or reduced neurological function. Prior ICU outcome studies have similarly shown that treatment intensity frequently reflects underlying severity, making ventilation-related variables useful predictors in mortality models [4], [7].

Vasopressor use is another major marker of high-risk status because these medications are commonly administered when blood pressure cannot be maintained adequately through fluids alone. Their use often indicates circulatory failure, septic shock, or profound hemodynamic instability. Wang et al. [11] reported that markers related to organ dysfunction and severe sepsis were strongly associated with mortality in critically ill patients, supporting the importance of hemodynamic support variables in predictive modeling.

Sepsis remains one of the most serious complications in intensive care because of its rapid systemic effects and potential progression to multiorgan failure. Shashikumar et al. [5] noted that early recognition of sepsis has become a major focus of machine learning research because delayed detection can significantly worsen outcomes. Liu et al. [8] also demonstrated the usefulness of predictive models for mortality estimation among ICU patients with sepsis.

Traditional severity measures continue to play an important role alongside treatment variables. APACHE and SOFA scores summarize illness burden and organ dysfunction using structured clinical criteria, which explains why they remain widely used in both conventional assessment and machine learning studies. Zhang et al. [3] showed that dynamic SOFA changes combined with age improved mortality prediction, reinforcing the relevance of severity scoring systems in modern risk models.

Age and comorbidity burden should also not be overlooked. Older patients and those with multiple chronic conditions may have reduced physiological reserve and lower tolerance to acute critical illness. These baseline vulnerability factors are repeatedly associated with poorer ICU outcomes and remain important complements to acute severity indicators [6], [9].

Overall, prior research suggests that high-risk ICU patients are usually identified through a combination of acute instability markers, severity scores, and baseline vulnerability factors rather than any single variable. This supports the use of multifactor predictive frameworks when estimating mortality risk in critical care settings [9].

D. AI in Critical Care

Artificial intelligence has become an increasingly important topic in critical care because intensive care units generate large volumes of structured and time-sensitive clinical data. Patient management often depends on rapid interpretation of multiple variables, including vital signs, laboratory trends, treatment interventions, and evolving organ dysfunction. In such settings, AI methods offer the potential to support clinicians by identifying hidden patterns, improving risk estimation, and assisting complex decision-making processes [9].

Recent literature suggests that AI applications in critical care extend beyond mortality prediction alone. Models have been developed for deterioration detection, sepsis screening, length-of-stay estimation, ventilator management, and treatment optimization. Sendak et al. [12] reviewed machine learning-enabled clinical decision support systems in ICUs and noted growing interest in tools that can assist clinicians through earlier warnings and more data-driven prioritization of care.

Despite this promise, successful implementation depends on more than predictive accuracy. Clinical environments require systems that are reliable, interpretable, and compatible with routine workflows. Komorowski et al. [9] emphasized that although AI can improve decision support, barriers such as

poor generalizability, missing data, workflow disruption, and limited trust continue to slow adoption in real healthcare settings.

Interpretability is especially important in ICU practice because high-stakes decisions often require clear justification. Clinicians may be less willing to rely on models that produce predictions without understandable reasoning. For this reason, many recent studies combine performance evaluation with feature importance analysis, transparent variables, or risk grouping methods that help explain how predictions are formed [6], [12].

Another practical consideration is that AI systems should complement rather than replace clinical judgment. ICU care involves contextual factors that may not be fully captured in structured datasets, including bedside examination findings, rapidly changing patient responses, and physician expertise. Predictive tools are therefore most useful when they support triage, prioritization, and earlier review instead of functioning as standalone decision makers.

Taken together, the literature indicates that effective ICU AI systems must combine technical performance with transparency and practical usability. Models that integrate smoothly into clinical workflow are more likely to deliver real value than systems evaluated only by headline metrics [9], [12].

E. Research Gap

Existing studies have shown that machine learning can contribute meaningfully to ICU mortality prediction, early warning systems, sepsis detection, and broader clinical decision support. However, the current literature also reveals several gaps that limit practical adoption and direct comparison across studies.

First, many published models were developed using highly specific patient cohorts, single-hospital datasets, or narrowly defined clinical scenarios. Examples include cardiac arrest populations, COVID-19 ICU cohorts, or diagnosis-specific mortality studies [2], [7], [8]. While these studies provide valuable findings, models trained under specialized conditions may not generalize well to broader ICU populations with mixed case complexity.

Second, several studies emphasize predictive performance metrics without giving equal attention to interpretability or operational usefulness. High accuracy or AUC values are important, but clinicians often need to understand why a patient is considered high risk before trusting model outputs in practice. Cabitza et al. [10] noted that lack of transparency and workflow integration remain key barriers to implementation in intensive care settings.

Another recurring issue is the treatment of imbalanced outcomes. In many ICU datasets, survivor cases outnumber mortality cases, which can lead models to favor the majority class. A system that achieves strong overall accuracy may still miss a substantial number of high-risk patients. For this reason, relying on accuracy alone may be insufficient when the practical goal is early recognition of deterioration or mortality risk.

High-dimensional feature sets are also common in healthcare modeling. Some studies rely on large numbers of predictors, many of which may be weak, redundant, or difficult to obtain consistently across institutions. This can reduce model transparency and complicate deployment, especially when simpler feature sets may achieve comparable usefulness [9].

The present study was designed to address these gaps through a practical and reproducible workflow. A public structured ICU dataset was used, feature refinement reduced predictors to seven clinically relevant variables, class imbalance was addressed, multiple models were compared under consistent conditions, and threshold tuning was used to improve recall-sensitive mortality detection. The framework also emphasized interpretability through feature importance analysis rather than relying only on a single performance metric.

IV. Comparative Analysis of Prior Studies

The reviewed studies demonstrate that ICU predictive analytics has been applied across different patient cohorts, datasets, and clinical objectives. However, direct numerical comparison remains difficult because results depend heavily on cohort design, feature availability, preprocessing methods, and outcome definitions. For this reason, comparisons are more meaningful when based on methodological patterns rather than raw metrics alone.

Several studies focused specifically on mortality prediction, while others emphasized early warning or deterioration detection. This suggests that ICU predictive analytics supports multiple clinical objectives rather than a single endpoint.

Another clear pattern is the tradeoff between technical performance and practical usability. More advanced ensemble methods may improve predictive power, but interpretable systems are often easier to trust and integrate into healthcare workflows [10].

The present study differs by combining feature refinement, class balancing, multi-model comparison, threshold tuning, and feature importance analysis within one reproducible end-to-end framework. Rather than prioritizing a single performance metric, the study emphasizes balanced and clinically usable prediction.

V. Methodology

This section describes the end-to-end workflow used to develop the ICU mortality prediction framework. The study followed a structured data science process beginning with dataset preparation, followed by exploratory analysis, feature refinement, model development, and performance evaluation. Additional steps were included to improve clinical usability, such as class balancing, decision-threshold tuning, and interpretation of final model outputs. The overall objective was to build a reproducible workflow that balances predictive performance with operational practicality.

The methodology was organized into the following stages: data acquisition, preprocessing, exploratory data analysis,

TABLE I
SUMMARY OF REPRESENTATIVE PRIOR STUDIES IN ICU PREDICTIVE ANALYTICS

Ref.	Study Focus	Cohort / Dataset	Method	Key Contribution	Main Limitation
[1]	Early warning system	Brazilian multicenter hospitals	ML models	External validation across hospitals	External performance may vary by institution
[2]	COVID ICU mortality	COVID ICU cohort	SuperLearner	Strong mortality prediction during pandemic	Disease-specific cohort
[3]	28-day mortality	ICU patients	ML + dynamic SOFA	Improved prediction using temporal SOFA + age	Focused outcome window
[4]	Mortality + LOS	Structured hospital ICU data	ML models	Dual prediction of mortality and stay length	Generalizability beyond local setting unclear
[7]	Cardiac arrest mortality	MIMIC-IV ICU cohort	ML models	Subgroup mortality prediction	Narrow subgroup
[8]	Sepsis mortality	ICU sepsis patients	ML models	Mortality prediction in severe infection	Specialized cohort
[10]	ICU AI barriers	Review study	Analysis	Highlighted trust and implementation barriers	No original predictive validation
[12]	ICU decision support	Literature review	Review	Summarized ML-enabled ICU support systems	No original predictive validation

statistical feature screening, balanced model training, comparative evaluation of multiple machine learning algorithms, and final interpretation through feature importance and risk stratification. Rather than relying on a single modeling step, the workflow was designed to examine how different decisions during the pipeline affect mortality prediction quality.

All experiments were implemented in Python using widely adopted data science libraries, including pandas and NumPy for data handling, matplotlib and seaborn for visualization, and scikit-learn for preprocessing, model training, and evaluation. XGBoost was additionally used for gradient boosting experiments because tree-based boosting methods have shown strong performance in structured tabular prediction tasks [6]. This software environment supports transparency, reproducibility, and future extension of the proposed framework.

To improve clarity, the methodology is divided into the following subsections: workflow overview, dataset information, data preparation, exploratory analysis, feature refinement and balancing, model development, threshold tuning, and model interpretation and practical evaluation.

A. Workflow Overview

The proposed workflow was designed as a complete data science pipeline for ICU mortality prediction using structured clinical data. Rather than treating modeling as a single isolated step, the process moved through successive stages that improved data quality, reduced unnecessary complexity, and supported more reliable evaluation.

The workflow began with data acquisition and initial inspection of the dataset. Variables were reviewed to understand their structure, data types, missing values, and suitability for predictive modeling. Identifier fields that did not provide clinical meaning were excluded to avoid introducing noise into the learning process.

The next stage focused on preprocessing and exploratory analysis. Numerical and categorical variables were examined through descriptive statistics, distribution plots, correlation analysis, and outcome-based comparisons. This step was used to identify clinically relevant patterns, detect imbalance in the mortality target, and understand relationships between candidate predictors and patient outcomes.

After exploration, feature refinement was performed to reduce the original predictor set into a smaller group of more informative variables. Statistical association testing, practical relevance, and interpretability were considered together during this stage. The final refined set contained seven predictors that were retained for downstream experiments.

Model development was then carried out using multiple supervised learning algorithms under consistent training and testing conditions. Because mortality cases were less frequent in the original dataset, class balancing was introduced during the final experiment to improve fairness of model learning. Comparative evaluation across models allowed stronger and weaker approaches to be identified using the same dataset conditions.

Following model comparison, decision-threshold tuning was applied to the strongest model so that mortality detection could be aligned more closely with the practical objective of minimizing missed high-risk patients. Final outputs were further interpreted through feature importance ranking, risk stratification analysis, and interpretation of final predictions.

A summary of the complete workflow is illustrated in Fig. 1, covering data preparation, feature refinement, class balancing, model development, threshold optimization, and robustness evaluation.

B. Dataset Information

The study used a publicly available Kaggle ICU patient monitoring and mortality prediction dataset containing 15,000 patient records and 24 variables relevant to mortality prediction. The dataset was selected because it provides a suitable environment for applying supervised machine learning methods to a realistic healthcare outcome problem while remaining accessible for reproducible academic experimentation.

Each record represented an ICU patient case and included demographic information, severity indicators, treatment variables, physiological summaries, laboratory-related measures,

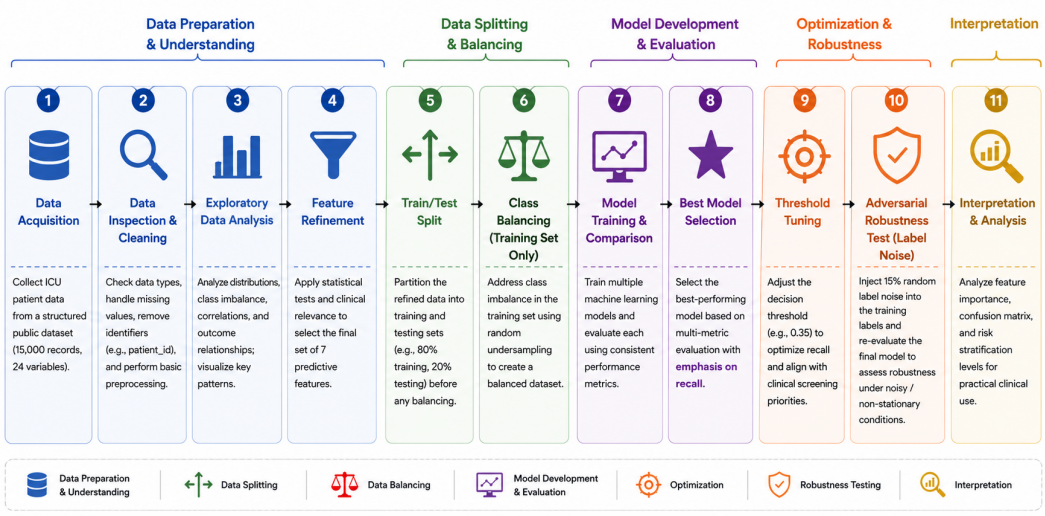

Fig. 1. End-to-end workflow of the proposed ICU mortality prediction framework.

and final outcome labels. The target variable for this study was mortality_label, a binary indicator representing survival or mortality outcome.

The original dataset included one identifier field, patient_id, which was excluded during preprocessing because patient identifiers do not provide predictive clinical meaning. The remaining variables were reviewed as candidate predictors for downstream modeling.

The available predictors included age, gender, admission type, comorbidity burden, APACHE score, SOFA score, heart-rate summaries, blood-pressure summaries, respiratory measures, oxygen saturation, temperature, glucose, lactate, urine output, ventilation requirement, vasopressor use, sepsis status, and length of stay. This combination of variables allowed the dataset to capture both baseline patient characteristics and indicators of acute physiological instability.

The dataset was considered appropriate for this project because it contains a mixture of interpretable clinical factors and measurable ICU interventions commonly associated with patient deterioration risk. At the same time, as with many public educational datasets, external clinical validation cannot be fully guaranteed. Findings should therefore be interpreted as part of a data science modeling study rather than a deployed medical prediction system.

C. Data Preparation

Before model development, the dataset was prepared to improve consistency, usability, and suitability for supervised learning. Data preparation focused on removing non-informative fields, checking variable structure, handling categorical inputs, and ensuring that predictors were ready for downstream experiments.

The identifier column, patient_id, was removed at an early stage because it represents a unique label rather than a clinically relevant predictor. Retaining identifier variables can introduce unnecessary noise and may distort model learning without improving predictive value.

The dataset was then inspected for data types, missing values, and general consistency. Numerical variables were retained in quantitative form, while categorical variables such as gender and admission type were encoded into machine-readable format during preprocessing pipelines. This allowed models to use both numeric and non-numeric information appropriately.

Before model training, predictor variables and the target outcome were separated. Feature matrices, X, contained all candidate predictors, while the response vector, y, contained the target variable, mortality_label. This structure supported consistent preprocessing, train-test splitting, and fair model evaluation.

Preprocessing steps used within modeling pipelines included median imputation for numerical variables where needed, standardization for algorithms sensitive to scale, and categorical encoding for non-numeric features. Applying these transformations inside pipelines helped prevent data leakage by ensuring that training information was learned only from the training subset.

Finally, the prepared dataset was reviewed for class distribution, which confirmed that survivor cases outnumbered mortality cases. This imbalance informed the later balancing strategy used in the final experiment. Overall, the preparation stage ensured that subsequent modeling was based on clean, structured data suitable for model evaluation.

D. Exploratory Analysis

Exploratory analysis was conducted to understand the structure of the dataset, identify important clinical patterns, and guide later modeling decisions. This stage focused on descriptive statistics, class distribution, variable relationships, and outcome-based comparisons using selected visualizations generated from the dataset.

Initial analysis showed that the target variable was imbalanced, with survivor cases substantially exceeding mortality cases, as illustrated in Fig. 2. Mortality cases therefore represented the minority class within the dataset. This finding was important because imbalanced outcomes can bias classifiers toward the majority class and reduce sensitivity to high-risk patients. The result directly motivated the balancing strategy used in the final experiment.

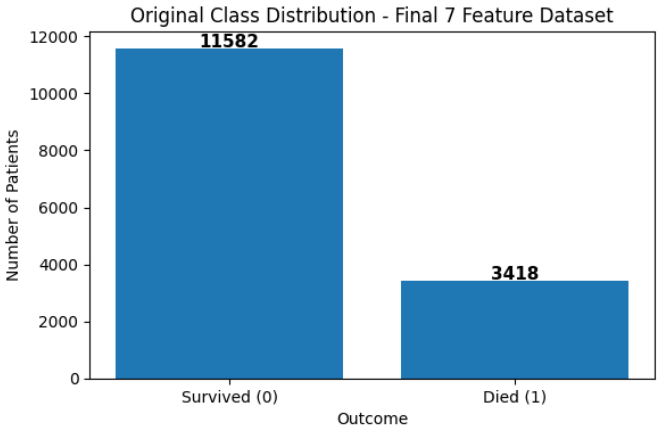

Fig. 2. Class distribution of mortality outcomes in the original dataset.

Several variables also showed clinically significant associations with mortality. Higher APACHE and SOFA scores were generally linked with worse outcomes, reflecting greater illness severity and organ dysfunction. Mortality rates were also higher among patients requiring ventilation support, as illustrated in Fig. 3, which is consistent with more severe ICU conditions.

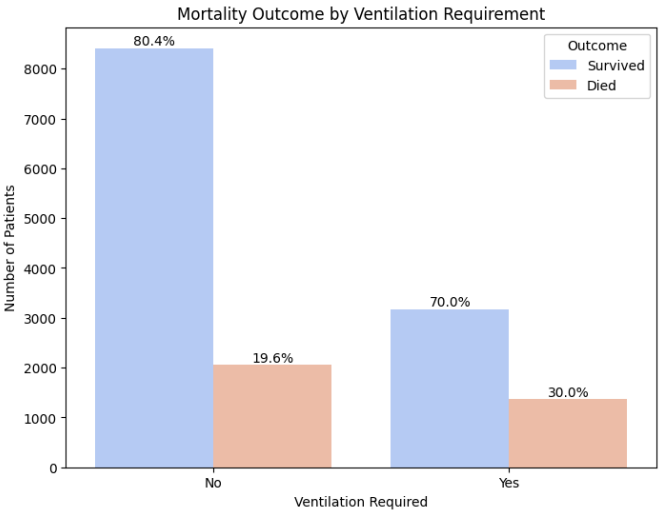

Fig. 3. Mortality outcome by ventilation support requirement in the original dataset.

Age-related trends were also observed. As shown in Fig. 4, mortality rates increased in older age groups, particularly among patients aged 61 years or older. This suggests that baseline vulnerability remained relevant alongside acute ICU conditions. In addition, sepsis status was associated with increased mortality, supporting prior findings that severe infection remains a major determinant of critical care outcomes [5], [8].

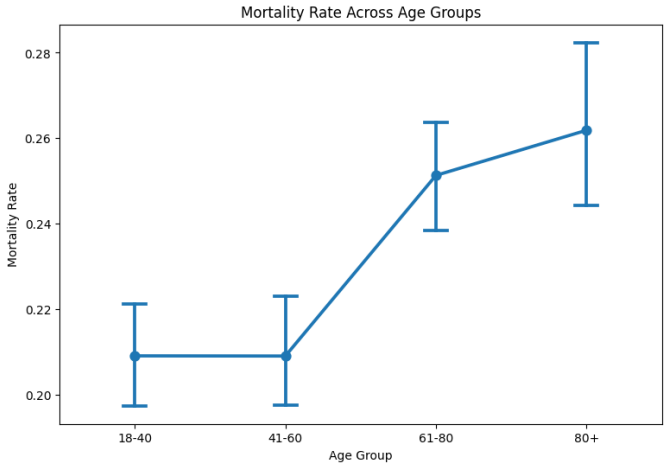

Fig. 4. Mortality rate across age groups in the original dataset.

Correlation screening and statistical testing were then used to examine relationships between candidate predictors and the target variable. Based on significance testing and practical relevance, seven predictors were retained for final modeling: ventilation requirement, vasopressor use, sepsis status, comorbidity score, SOFA score, age, and APACHE score. The strongest statistically supported variables are summarized in Table II.

TABLE II
TOP STATISTICALLY ASSOCIATED PREDICTORS RETAINED FOR FINAL MODELING

Rank	Feature	Correlation (r)	P-Value
1	ventilation_required	0.1139	< 0.001
2	vasopressor_used	0.1042	< 0.001
3	sepsis_flag	0.0926	< 0.001
4	apache_score	0.0653	< 0.001
5	comorbidity_score	0.0636	< 0.001
6	sofa_score	0.0631	< 0.001
7	age	0.0587	< 0.001

These exploratory findings provided practical justification for subsequent feature refinement and model development stages.

E. Feature Refinement and Balancing

Following exploratory analysis, a refinement stage was conducted to reduce the original predictor set into a smaller and more informative group of variables for final modeling. Although the dataset contained a broad range of features, not all predictors were equally informative, and some showed weak association with mortality or limited added value. Reducing unnecessary inputs can improve interpretability, simplify model training, and lower the risk of overfitting.

Feature selection was guided by a combination of statistical evidence and clinical relevance. Correlation screening, significance testing, observed outcome trends, and domain interpretability were considered together rather than relying on a single automated criterion. Based on this process, seven predictors were retained for the final modeling stage: ventilation requirement, vasopressor use, sepsis status, APACHE score, SOFA score, comorbidity score, and age.

These retained variables reflect a combination of acute physiological severity, treatment intensity, infection status, chronic disease burden, and baseline patient vulnerability. The reduced feature set was considered more efficient than using the full original variable list while still preserving valuable information for mortality prediction.

A second challenge identified during exploratory analysis was class imbalance. Survivor cases substantially outnumbered mortality cases in the original dataset, which can bias models toward predicting the majority outcome. In healthcare prediction tasks, this issue is especially important because missed mortality cases may reduce the usefulness of a screening system.

To address this problem, random undersampling of the majority class was applied during the final training stage to create a more even class distribution. This strategy was selected to reduce majority-class bias during model learning and strengthen sensitivity to mortality cases without introducing synthetic observations. The resampled dataset was then used for comparative model training under consistent experimental conditions.

This refinement and balancing stage was intended to strengthen interpretability and improve predictive usefulness before final model development.

F. Model Development

After data preparation, feature refinement, and class balancing, multiple supervised machine learning models were developed for binary mortality prediction. Evaluating several approaches was important because different classifiers vary in their ability to capture nonlinear relationships, separate decision boundaries, and model feature interactions within healthcare data.

The selected models included Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, Gradient Boosting, and XGBoost. This set was chosen to provide methodological diversity by representing linear models, distance-based methods, probabilistic classifiers, tree-based learners, and ensemble techniques.

All models were trained under the same experimental framework to support fair comparison. The refined seven-feature dataset was divided into training and testing subsets, and identical evaluation procedures were applied across classifiers. Performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC rather than relying on a single metric.

Particular emphasis was placed on recall because the main objective of the project was to improve identification of high-risk mortality cases. In healthcare prediction tasks, a model with moderate overall accuracy may still be more useful than a higher-accuracy alternative if it correctly detects a larger proportion of true mortality outcomes.

Initial baseline experiments were followed by a final balanced training stage using the refined predictor set. This progression allowed comparison between broader early experiments and the final optimized framework. The strongest-performing models were then carried forward for threshold tuning and detailed interpretation in the results section.

This model development phase was intended not only to compare algorithms, but also to determine which learning approach was most suitable for ICU mortality risk screening.

G. Threshold Tuning

After comparative model evaluation, threshold tuning was applied to the best-performing model in order to align predictions more closely with the project objective of mortality risk detection. Many binary classification models generate predicted probabilities, which are commonly converted into class labels using a default threshold of 0.50. However, this default setting is not always optimal, particularly in healthcare tasks where missing high-risk cases may carry greater consequences than generating additional false alarms.

Based on the comparative experiments, XGBoost was selected for further refinement because it showed the most suitable balance of classification performance and mortality-case detection. The probability threshold was then adjusted below the default value and evaluated at several candidate levels, including 0.35, 0.40, and 0.45.

Lowering the threshold increases the likelihood that uncertain cases are classified as mortality risk, which typically improves recall while reducing precision. This tradeoff was considered appropriate for the present study because identifying a larger proportion of true high-risk patients was prioritized over maximizing overall accuracy alone.

Among the tested settings, a threshold of 0.35 provided the most suitable balance for the study objective. At this level, the model achieved the highest recall among tested thresholds while maintaining acceptable F1-score and ROC-AUC performance. For this reason, the 0.35 threshold was selected as the final operating point for detailed evaluation in the results section.

Threshold tuning showed that model performance also depends on how predicted probabilities are converted into final labels.

In addition to threshold tuning, a robustness evaluation was conducted by introducing controlled label noise into the training data. This step simulates potential inconsistencies in real-world clinical datasets and evaluates how model performance changes under noisy conditions, particularly in terms of mortality detection.

VI. Discussion and Results

The experimental findings showed notable differences across the evaluated models and highlighted how metric selection can influence interpretation of predictive performance.

Because ICU mortality prediction involves both overall classification accuracy and identification of high-risk patients, results were examined using multiple evaluation measures rather than a single score.

The study progressed through three experimental stages, beginning with broader baseline comparisons and concluding with the strongest overall framework in Experiment 3. This staged approach allowed earlier findings to guide later improvements in feature refinement, class balancing, and threshold optimization.

Comparative experiments indicated that model choice had a clear effect on classification behavior. Some algorithms achieved slightly stronger overall accuracy, whereas others demonstrated better sensitivity to mortality cases. This distinction is especially important in healthcare settings, where failure to identify high-risk patients may carry greater consequences than a modest reduction in overall accuracy.

Among the tested approaches, tree-based ensemble methods produced the most competitive overall performance, particularly XGBoost and Random Forest. Simpler methods such as Logistic Regression and Naive Bayes remained competitive in selected metrics, while single Decision Tree performance was comparatively weaker. These findings are consistent with broader machine learning literature, where ensemble methods often perform strongly on structured tabular data [6], [9].

Because the main objective of the project was improved identification of mortality cases, additional discussion focuses on recall, threshold behavior, and model interpretability rather than accuracy alone. The final selected model was therefore determined not only by raw metrics, but also by suitability for the intended screening objective.

The following subsections examine model comparison results, final model selection, threshold tradeoff analysis, feature importance, confusion matrix interpretation, and comparison with prior studies.

A. Model Comparison Results

Model comparison was conducted to evaluate how different machine learning approaches performed under the final experimental framework. The study progressed through three stages of experimentation, with earlier phases used for baseline comparison and feature refinement, while Experiment 3 represented the final optimized setting that combined the refined seven-feature dataset, class balancing, and threshold evaluation. Accordingly, the results discussed in this section primarily emphasize Experiment 3.

Clear performance differences were observed across the evaluated models. As summarized in Table III, ensemble methods generally produced the most competitive multi-metric results, particularly XGBoost, Random Forest, and Gradient Boosting. These approaches tended to balance recall, precision, and ROC-AUC more effectively than simpler single-model methods.

Logistic Regression and Naive Bayes remained reasonably competitive, especially in selected metrics such as accuracy and precision. This suggests that simpler models can still provide useful baselines when interpretability or computational simplicity is preferred. However, their sensitivity to mortality cases was lower than the strongest ensemble approaches.

Support Vector Machine demonstrated stable mid-range performance, while K-Nearest Neighbors and Decision Tree produced comparatively weaker overall results. In particular, the single Decision Tree model showed the lowest combined performance across several metrics, which is consistent with its greater susceptibility to unstable splits and overfitting on structured tabular data.

Among all evaluated settings, the tuned XGBoost model using a threshold of 0.35 showed the most suitable performance for the project objective because it combined very high recall with acceptable precision and F1-score. Although its raw accuracy was not the highest, it was better aligned with the goal of improving detection of mortality cases.

These findings indicate that model comparison should not rely on accuracy alone. In healthcare screening problems, stronger recall and balanced metric performance may be more valuable than modest gains in overall classification accuracy.

B. Final Model Selection

Although several models produced competitive results, final model selection was based on suitability for the project objective rather than a single performance metric. The main aim of the study was to improve identification of mortality cases, which made sensitivity to high-risk outcomes more important than maximizing overall accuracy alone.

Among all evaluated approaches, the tuned XGBoost model with a probability threshold of 0.35 was selected as the final model. As shown in Table III, this setting achieved a recall of 0.885, indicating that a large proportion of mortality cases were correctly identified. It also maintained an F1-score of 0.678 and ROC-AUC of 0.641, demonstrating a reasonable balance between case detection and overall discrimination.

Some competing models achieved slightly higher accuracy, particularly Logistic Regression (0.607) and Naive Bayes (0.604). However, these models showed substantially lower recall values of 0.561 and 0.575, respectively. In the context of ICU mortality screening, higher accuracy with reduced mortality detection was considered less desirable than stronger recall with moderate accuracy.

The selected XGBoost model also outperformed the untuned XGBoost baseline, demonstrating that threshold adjustment substantially improved screening performance. Whereas the default threshold version produced a recall of 0.607, lowering the threshold to 0.35 increased recall to 0.885, with an accompanying improvement in F1-score.

XGBoost likely performed strongly because boosting methods can capture nonlinear relationships and interactions among clinical predictors while remaining effective on structured tabular datasets. Another advantage of the selected model was its support for later interpretation steps, including feature importance ranking and risk stratification analysis. This made the final model useful not only for prediction, but also for

TABLE III
COMPARATIVE PERFORMANCE OF EVALUATED MODELS IN EXPERIMENT 3

Model	Threshold	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Tuned XGBoost	0.35	0.580	0.550	0.885	0.678	0.641
Tuned XGBoost	0.40	0.588	0.562	0.800	0.660	0.641
Tuned XGBoost	0.45	0.602	0.584	0.709	0.640	0.641
SVM	0.50	0.597	0.591	0.626	0.608	0.634
Random Forest	0.50	0.601	0.598	0.617	0.607	0.641
XGBoost	0.50	0.599	0.597	0.607	0.602	0.641
Gradient Boosting	0.50	0.598	0.597	0.604	0.600	0.649
Naive Bayes	0.50	0.604	0.610	0.575	0.592	0.646
Logistic Regression	0.50	0.607	0.618	0.561	0.589	0.641
KNN	0.50	0.567	0.566	0.573	0.570	0.582
Decision Tree	0.50	0.577	0.583	0.541	0.561	0.613

explaining which clinical factors contributed most strongly to estimated risk.

The tuned XGBoost model was therefore selected because it best matched the intended use case of mortality risk screening, where minimizing missed high-risk patients was prioritized over modest gains in overall accuracy.

TABLE IV
THRESHOLD PERFORMANCE OF TUNED XGBOOST

Threshold	Acc.	Prec.	Recall	F1	ROC-AUC
0.35	0.580	0.550	0.885	0.678	0.641
0.40	0.588	0.562	0.800	0.660	0.641
0.45	0.602	0.584	0.709	0.640	0.641

C. Threshold Tradeoff Analysis

Threshold tuning showed that classification performance changed noticeably as the decision threshold was adjusted below the default value of 0.50. This confirms that model performance depends not only on the selected algorithm, but also on how predicted probabilities are converted into final class labels.

As summarized in Table IV, lowering the threshold from 0.45 to 0.40 and then to 0.35 progressively increased recall from 0.709 to 0.800 and finally 0.885. This indicates that a larger proportion of true mortality cases were detected as the threshold became less restrictive.

The improvement in recall was accompanied by expected tradeoffs in other metrics. Precision declined from 0.584 at the 0.45 threshold to 0.562 at 0.40 and 0.550 at 0.35, reflecting a higher number of false-positive alerts. Accuracy also decreased modestly as recall increased. These changes are consistent with standard binary classification behavior, where greater sensitivity often comes at the cost of stricter precision.

From a practical perspective, this tradeoff was considered acceptable for the present study. In ICU mortality screening, failing to identify a genuinely high-risk patient may carry greater consequences than flagging some additional lower-risk cases for review. For this reason, recall was given stronger weight than raw accuracy alone.

Although the 0.45 threshold produced a more balanced overall metric profile, the 0.35 threshold provided the strongest alignment with the project objective of minimizing missed mortality cases. It also preserved the highest F1-score among the tested threshold settings, supporting its selection as the final operating point.

These results indicate that threshold selection is an important modeling decision rather than a fixed default setting.

D. Feature Importance

Interpretability was considered an essential component of the final modeling framework because healthcare prediction systems are often more useful when clinicians can understand which variables most strongly influence estimated risk. For this reason, feature importance analysis was conducted using the final tuned XGBoost model.

As shown in Fig. 5, treatment-related indicators were the strongest contributors to model predictions. Vasopressor use ranked as the most influential feature, followed closely by ventilation requirement and sepsis status. These findings are clinically reasonable because circulatory instability, respiratory support needs, and severe infection are commonly associated with critical deterioration and mortality risk in ICU settings.

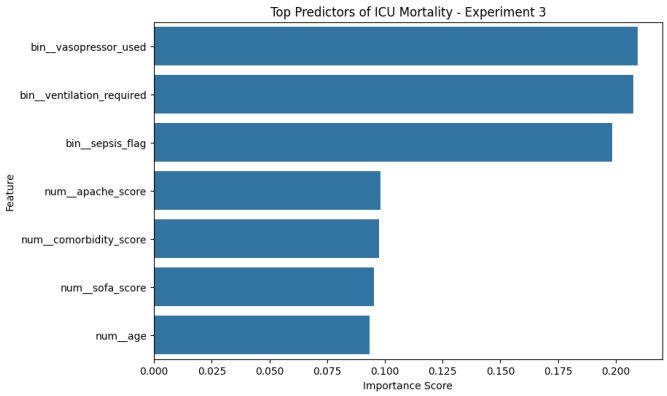

Fig. 5. Relative feature importance of predictors in the final tuned XGBoost model.

Traditional severity measures also remained influential. APACHE score, comorbidity score, and SOFA score each contributed considerably to model decisions, indicating that

structured severity assessment still provides useful information when combined with machine learning methods.

Age was the lowest-ranked predictor among the final seven variables, but it continued to provide additional predictive value. This suggests that baseline vulnerability remained relevant, although acute illness markers and treatment intensity carried stronger influence within the final model.

The overall ranking pattern indicates that the selected model relied on a combination of acute clinical instability, chronic health burden, and demographic vulnerability rather than any single predictor alone. This supports the broader view that ICU mortality risk is multifactorial and best assessed through integrated indicators.

Feature importance analysis does not imply direct causation. Instead, it provides insight into which variables were most useful for prediction within the studied dataset. Nevertheless, these findings suggest that the final model was relying on clinically relevant information rather than arbitrary patterns.

E. Confusion Matrix Interpretation

To examine prediction behavior beyond summary metrics, a confusion matrix was reviewed for the final tuned XGBoost model using the selected threshold of 0.35. This analysis provides a clearer view of how many cases were correctly or incorrectly classified within each outcome category.

As shown in Fig. 6, the model correctly identified 605 mortality cases as mortality-risk predictions, while 79 mortality cases were missed and classified as survivors. This result is consistent with the high recall value reported earlier and supports the use of the selected threshold for improved mortality-case detection.

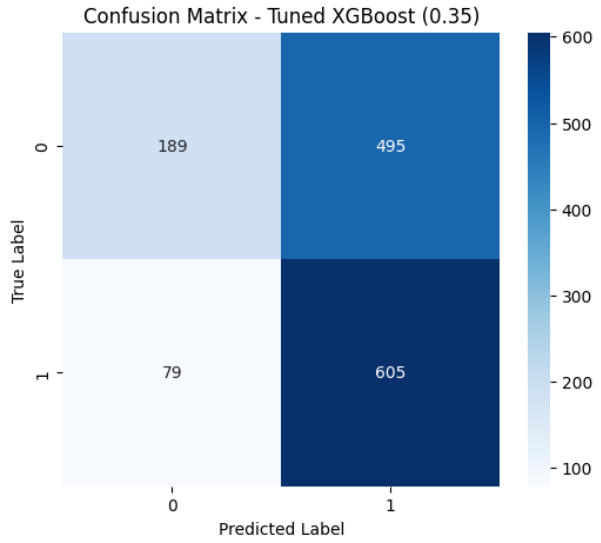

Fig. 6. Confusion matrix of the final tuned XGBoost model at a probability threshold of 0.35.

For survivor outcomes, the model correctly classified 189 cases, while 495 survivor cases were incorrectly flagged as mortality risk. Although this increased the number of false-positive alerts, it was an expected consequence of lowering the threshold to prioritize sensitivity.

From a practical perspective, the confusion matrix reflects the central tradeoff of the study. The final model was intentionally calibrated to reduce false negatives, meaning fewer high-risk patients would be missed. In ICU screening contexts, this may be preferable to a stricter model that misses more mortality cases while producing fewer alerts.

At the same time, the relatively high false-positive count suggests that model outputs should be interpreted as screening support rather than definitive clinical judgment. Flagged patients would still require human review and clinical context before action is taken.

The confusion matrix therefore supports the decision to prioritize recall-focused classification. The model favored broader detection of mortality risk while preserving useful overall predictive performance.

F. Comparison with Prior Studies

The findings of this study were generally consistent with earlier research showing that machine learning can support ICU mortality prediction using structured clinical data. As reported in prior studies, model performance differed across algorithms, feature selection choices, class balance methods, and evaluation criteria [4], [7], [12].

In the current experiments, tree-based ensemble methods such as XGBoost and Random Forest produced the strongest overall results. Similar patterns have been reported in previous ICU prediction studies, where ensemble models often performed well on tabular healthcare data because they can capture nonlinear relationships between predictors [2], [4].

The results also showed that traditional severity indicators remained useful within data-driven models. APACHE score, SOFA score, comorbidity burden, and age all contributed meaningfully in the feature importance analysis. This is consistent with earlier studies linking illness severity and baseline vulnerability to ICU outcomes [3], [9].

One difference in the present framework was the stronger emphasis on threshold tuning and recall-focused model selection. Many published studies mainly report accuracy or ROC-AUC values, whereas this project prioritized improved detection of mortality cases. From a screening perspective, missing fewer high-risk cases may be more valuable than achieving slightly higher overall accuracy.

Another difference was the use of a reduced seven-feature framework instead of a high-dimensional predictor set. While some studies rely on extensive variable lists, the current findings suggest that useful and interpretable performance can still be achieved with a smaller group of clinically relevant predictors.

Direct numerical comparison with prior studies should be made carefully because datasets, patient populations, mortality definitions, and preprocessing methods differ across publications. Nevertheless, the results support the growing

role of interpretable machine learning methods in ICU risk stratification and clinical decision-support applications.

VII. Limitations and Future Work

Although the study produced promising findings, several limitations should be acknowledged. The analysis was conducted using a publicly available secondary dataset rather than data collected directly from a hospital environment. While this supports reproducibility and academic experimentation, external clinical validity cannot be fully guaranteed. Model performance may differ when applied to real-world ICU populations with different demographics, treatment protocols, and documentation practices.In addition, performance estimates were based on a single dataset, so results may differ when applied to external hospitals or different ICU populations.

The dataset also contained structured summary variables rather than detailed longitudinal clinical records. ICU outcomes often evolve over time, and time-series information such as changing vital signs, laboratory trends, or treatment response may improve prediction compared with static inputs alone. Future studies could evaluate sequential modeling approaches using richer temporal data.

Another limitation concerns class balancing. Random undersampling was selected to reduce majority-class bias and improve mortality-case detection. However, undersampling removes a portion of available survivor cases, which may discard potentially useful information. Alternative strategies such as class-weighted learning, SMOTE-based resampling, or cost-sensitive optimization could be explored in future work.

The study also emphasized interpretability through a reduced seven-feature framework. While this simplified modeling and improved transparency, excluding additional variables may have limited peak predictive performance. Future research may compare compact interpretable models with larger feature sets under the same evaluation framework.

In addition, the final model was optimized for recall-focused screening rather than definitive clinical diagnosis. Predictions should therefore be interpreted as decision-support signals rather than standalone medical judgments. Real deployment would require prospective validation, clinician oversight, calibration testing, and ethical review before operational use.

Future work may also extend the framework through external validation on multi-center ICU datasets, deeper hyperparameter optimization, calibration analysis, and integration into practical triage-support systems. These directions would help determine whether the current experimental framework can translate into robust clinical decision-support performance.

VIII. Conclusion

This study developed and evaluated a practical data science framework for ICU mortality prediction using structured clinical data. The workflow combined data preparation, exploratory analysis, feature refinement, class balancing, comparative model evaluation, and threshold tuning within a reproducible experimental process.

The results showed that model choice had a meaningful effect on predictive performance, with tree-based ensemble methods producing the most competitive overall outcomes. Among the evaluated approaches, the tuned XGBoost model using a probability threshold of 0.35 was selected as the final model because it best matched the project objective of improved mortality-case detection. This setting achieved a recall of 0.885 while maintaining acceptable F1-score and ROC-AUC performance.

Feature importance analysis further indicated that vasopressor use, ventilation requirement, sepsis status, APACHE score, SOFA score, comorbidity burden, and age were the most relevant predictors within the final framework. These findings support the view that ICU mortality risk is influenced by a combination of acute instability, chronic health burden, and baseline vulnerability.

The results of this study also showed that healthcare prediction models should not be judged by accuracy alone. In screening-oriented applications, threshold selection, recall performance, and interpretability may be equally important when assessing real-world clinical utility.

Although additional validation is required before clinical deployment, the findings suggest that interpretable machine learning methods can provide valuable support for ICU risk stratification and future decision-support systems.

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